4 A rigid flat plate is made to vibrate vertically with simple harmonic motion. The frequency of the vibration is controlled by a signal generator as shown in Fig. 4.1.

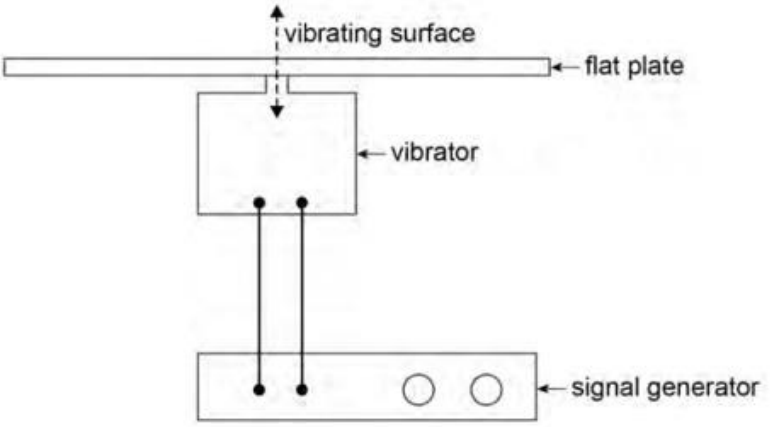

Fig. 4.1

Taking upward direction as positive, the variation with time *t* of the velocity *v* for the vibrating plate at one frequency is shown in Fig. 4.2.

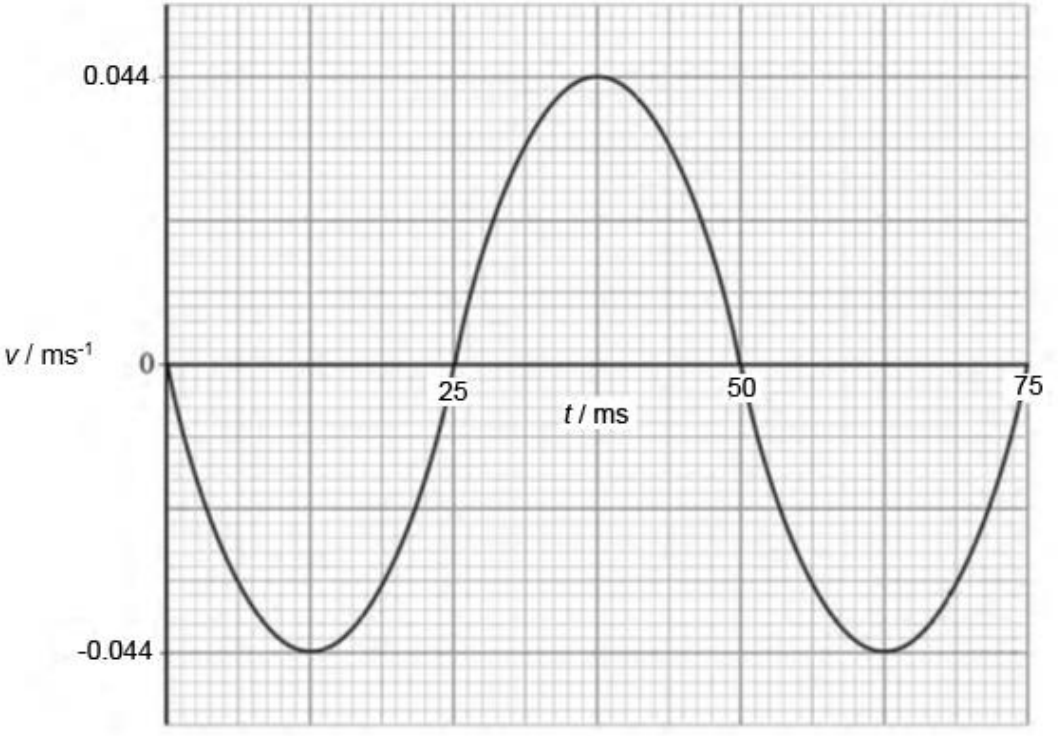

Fig. 4.2

(a) Show that the maximum displacement of the plate is 3.5×10^{-4} m.

[1]

(b) Draw on Fig. 4.3 the variation with time *t* of the displacement *s* of the plate between 0 and 75 ms.

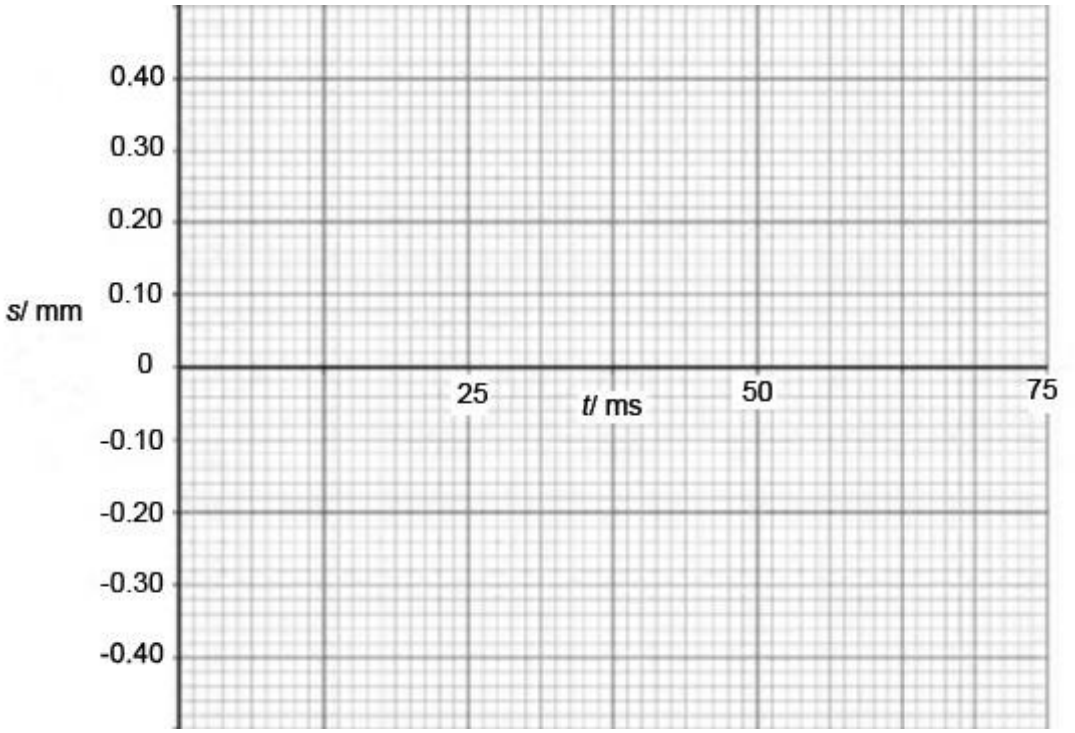

Fig. 4.3 [2]

(c) State **one** time at which the plate has maximum potential energy.

time = ... s [1]

(d) A small quantity of fine sand is placed onto the surface of the plate. Initially the sand grains stay in contact with the plate as it vibrates. The frequency of the vibrator is then gradually increased. Above a particular frequency the sand grains lose contact with the surface.

Explain how and why this happens.

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..... [3]